COLIMITS OF TOPOLOGICAL GROUPS DIFFERING IN TOPOLOGICAL SPACES AND TOPOLOGICAL GROUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The step 'it is easy to show that U cap G_n is open' is the author's own terse carried claim (6895-6896), used verbatim; the irrationality of pi supplies the never-integral-multiple fact.

Editorial note. The label lemma-colimit-topology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 6879-6887: G_n = Q x R^n as a topological group with the Euclidean topology, closed embeddings G_n -> G_{n+1} appending a zero; G = colim G_n with the colimit topology (U open iff G_n cap U open for all n); the claim is that this is not a topological group
- Context span 6889-6909: the set U = {(x_0, x_1, ...) : |x_j| < |cos(j x_0)| for j > 0}; since j x_0 is never an integral multiple of pi/2 (pi irrational) it is easy to show U cap G_n is open; if the colimit topology made G a topological group there would be an open neighbourhood V of 0 with V + V subset U; then for every j >= 0 some epsilon_j > 0 has (0, ..., 0, x_j, 0, ...) in V for |x_j| < epsilon_j; V + V subset U would force (x_0, 0, ..., 0, x_j, 0, ...) in U for |x_0| < epsilon_0 and |x_j| < epsilon_j; taking j with j epsilon_0 > pi/2 and x_0 in Q with |cos(j x_0)| < epsilon_j gives |cos(j x_0)| < |x_j| < epsilon_j, contradicting membership in U

$G_n = \mathbf{Q} \times \mathbf{R}^n$, $n \geq 1$ seen as a topological group for addition endowed with the usual (Euclidean) topology. Consider the closed embeddings $G_n \to G_{n+1}$ mapping $(x_0, \ldots, x_n)$ to $(x_0, \ldots, x_n, 0)$. We claim that $G = \operatorname{colim} G_n$ endowed with the topology

$$U \subset G \text{ open} \Leftrightarrow G_n \cap U \text{ open } \forall n$$

is not a topological group.

To see this we consider the set

$$U = \{(x_0, x_1, x_2, \ldots) \text{ such that } |x_j| < |\cos(jx_0)| \text{ for } j > 0\}$$

Using that jx_0 is never an integral multiple of $\pi/2$ as π is not rational it is easy to show that $U \cap G_n$ is open. Since $0 \in U$, if the topology above made G into a topological group, then there would be an open neighbourhood $V \subset G$ of 0 such

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that $V + V \subset U$. Then, for every $j \geq 0$ there would exist $\epsilon_j > 0$ such that $(0, \ldots, 0, x_j, 0, \ldots) \in V$ for $|x_j| < \epsilon_j$. Since $V + V \subset U$ we would have

$$(x_0, 0, \ldots, 0, x_j, 0, \ldots) \in U$$

for $|x_0| < \epsilon_0$ and $|x_j| < \epsilon_j$. However, if we take j large enough such that $j\epsilon_0 > \pi/2$, then we can choose $x_0 \in \mathbf{Q}$ such that $|\cos(jx_0)|$ is smaller than ϵ_j, hence there exists an x_j with $|\cos(jx_0)| < |x_j| < \epsilon_j$. This contradiction proves the claim.

Lemma 0.1. *There exists a system $G_1 \to G_2 \to G_3 \to \ldots$ of (abelian) topological groups such that $\operatorname{colim} G_n$ taken in the category of topological spaces is different from $\operatorname{colim} G_n$ taken in the category of topological groups.*

Proof. See discussion above. □